

PROJECT DESIGN PHASE

Team ID : NM2025TMID03763

Project Title : Calculating Family Expenses using ServiceNow Administrator

Performance Testing Phase: “Calculating Family Expenses Using ServiceNow”

1. Purpose

The Performance Testing phase aims to verify that the ServiceNow-based family expense management system meets defined performance benchmarks for speed, scalability, and reliability. It evaluates how the application behaves under normal and peak usage conditions.

2. Objectives

- Assess system response time for critical operations (expense submission, report generation, dashboard load).
- Identify and eliminate potential performance bottlenecks.
- Validate that the system supports expected user load without degradation.
- Ensure the system can recover gracefully from high-traffic scenarios.
- Establish baseline performance metrics for future improvements.

3. Scope

In Scope

- Load and stress testing of ServiceNow workflows and forms.
- Performance validation of dashboards and reports.
- Response time measurement for expense creation, updates, and queries.
- System scalability analysis for different user loads.

Out of Scope

- External API performance (to be tested during integration phase).
- Network-level and infrastructure-level testing (handled by ServiceNow platform).

4. Performance Metrics

Metric	Target	Description
Response Time	≤ 3 seconds	Time to load dashboards or submit expenses
Throughput	≥ 50 transactions/min	Number of expense submissions processed per minute
CPU Utilization	$\leq 70\%$	Resource consumption under normal load
Memory Usage	$\leq 65\%$	Memory footprint during peak load
Error Rate	$\leq 1\%$	Ratio of failed transactions to total requests
Concurrent Users	Up to 10 family members	Simulated concurrent access by all users

Availability	99% uptime	Application accessibility during testing window
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5. Test Environment Setup

Component	Configuration
Platform	ServiceNow Personal/Dev Instance
Modules	Custom Tables, Flow Designer, Dashboards
Tools Used	ServiceNow ATF (Automated Test Framework), JMeter (optional for load simulation)
Test Data	500–1000 expense records, multiple categories, and budget entries
Network	Standard broadband connection (simulate real-world latency)

6. Test Scenarios

ID	Scenario	Objective	Expected Result
PT1	Submit expense entry	Validate form submission response time	< 3 seconds
PT2	Load Expense Dashboard	Test dashboard data rendering time	< 3 seconds
PT3	Run Monthly Report	Assess report generation speed	< 5 seconds
PT4	Trigger Budget Alert Workflow	Validate automation under load	Workflow completes < 2 sec

PT5	10 concurrent users submitting expenses	Test concurrency and stability	No degradation or errors
PT6	Upload bulk data (1000 records)	Evaluate database write performance	Successful load within limits
PT7	Simulate 1-hour peak usage	Measure sustained throughput	System remains stable

7. Testing Types

- **Load Testing:** Validate performance under expected user loads.
- **Stress Testing:** Determine system behavior under extreme or breaking loads.
- **Endurance Testing:** Assess performance over prolonged operation (8–12 hours).
- **Spike Testing:** Test sudden increases in user activity (e.g., month-end spending).
- **Scalability Testing:** Evaluate how system performance scales with more users or data.

8. Performance Testing Process

1. **Planning:** Define key performance metrics and success criteria.
2. **Environment Setup:** Prepare ServiceNow instance and test data.
3. **Test Script Development:** Create automated and manual test scripts in ATF or JMeter.
4. **Execution:** Perform load and stress tests under varying conditions.

5. **Monitoring:** Track CPU, memory, transaction rates, and response times.
6. **Analysis:** Identify bottlenecks, query inefficiencies, or slow workflows.
7. **Optimization:** Apply improvements (indexing, workflow tuning, caching).
8. **Re-testing:** Validate that optimizations meet defined targets.
9. **Reporting:** Document results, charts, and recommendations.

9. Roles & Responsibilities

Role	Responsibility
Performance Test Lead	Define test plan and metrics
ServiceNow Developer	Optimize workflows and data models
Tester / QA Engineer	Execute tests and log results
System Administrator	Monitor instance performance and logs
Project Manager	Track testing progress and approve reports

10. Risk and Mitigation

Risk	Impact	Mitigation
Slow dashboard rendering	High	Optimize queries and reduce widgets

Workflow delays	Medium	Streamline triggers and conditions
Data volume impact	Medium	Use indexing and table cleanup strategies
Server throttling in dev instance	Low	Schedule tests during off-peak hours
Insufficient test data	Low	Generate synthetic data sets

11. Success Criteria

- All critical transactions execute within 3 seconds.
- No workflow or form errors under 10 concurrent users.
- Dashboards and reports load within acceptable limits.
- No data loss or corruption during heavy load tests.
- Post-optimization performance improvement $\geq 20\%$.

12. Deliverables

- Performance Test Plan
- Test Scripts (ATF or JMeter)
- Test Execution Reports (Response Time, Throughput)
- Bottleneck Analysis Report
- Optimization Recommendations
- Final Performance Certification Report

13. Next Steps

1. Conduct baseline test with sample data.
2. Tune workflows and forms based on results.
3. Perform final load and stress testing.
4. Obtain stakeholder sign-off for production readiness.
5. Move to **User Acceptance Testing (UAT) Phase.**